

Real-Time Object Detection & Counting System

📌 Overview

This project uses computer vision to detect and count objects in real time using a webcam. It is built using YOLOv8 and OpenCV.

🚀 Features

- * Real-time object detection
- * Object counting
- * Live webcam feed
- * Bounding box visualization

📁 Tech Stack

- * Python
- * OpenCV
- * YOLOv8 (Ultralytics)

⚙️ Setup Instructions

1. Clone the repository:

```
git clone <https://github.com/sagarsinghrajput0226/byop_project_23bai10299.git>
```

2. Navigate to folder:

```
cd BYOP_Object_Detection
```

3. Install dependencies:

```
pip install -r requirements.txt
```

4. Run the project:

```
python app.py
```

🎯 Usage

- * The webcam will open automatically
- * Objects will be detected and counted in real time
- * Press ESC to exit

📊 Applications

- * Crowd monitoring
- * Retail analytics
- * Security systems

🧑 Author

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